

BorderWatch: Intelligent Multi-Station Surveillance and Object Correlation System

A CAPSTONE PROJECT REPORT

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Artificial Intelligence and Data Science

Submitted by

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DECLARATION

We M.V. Priyavarsheni, S. Ashira Bridget Susan of Artificial Intelligence and Data Science, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled “BorderWatch: Intelligent Multi-Station Surveillance and Object Correlation System” is the result of our own Bonafide efforts.

To the best of our knowledge, the work presented here is original, accurate and has been carried out in accordance with principles of the engineering ethics.

Place : Chennai

Date: 18/08/2026

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “BorderWatch: Intelligent Multi-Station Surveillance and Object Correlation System” is the bonafide work of M.V. Priyavarsheni (192524428), S. Ashira Bridget Susan (192524316) under the supervision of Dr. C. Pushpalatha and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech computer science and engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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Abstract

Border and perimeter security operations depend on continuous visual monitoring across many camera stations spread over wide and often remote geographic areas, yet conventional CCTV setups leave this monitoring largely manual, station-isolated, and reactive. This report presents BorderWatch, an intelligent multi-station surveillance and object correlation system that automatically detects, tracks, and correlates objects of interest — people, vehicles, and other assets — as they move across the fields of view of multiple, geographically distributed camera stations. BorderWatch combines real-time object detection, single-station multi-object tracking, and a cross-station object correlation engine that uses appearance features, motion trajectories, and timing information to re-identify the same object as it reappears at a different station, producing a unified movement record instead of a set of disconnected clips. In this study, each station is modelled as a node that streams detections and track metadata to a central correlation service, which matches candidate objects across stations, resolves ambiguous matches, and raises rule-based alerts for suspicious or unauthorized movement patterns. By analysing recorded and simulated multi-camera footage, the system is evaluated for detection accuracy, cross-station re-identification accuracy, correlation latency, and false-alarm rate. The performance of the approach is evaluated based on detection precision and recall, identity-switch rate, average correlation time between stations, and alert response time. The results demonstrate that combining object detection with cross-station correlation significantly improves situational awareness and reduces the manual effort required to trace an object's movement through a multi-station network, while highlighting trade-offs between correlation strictness and false-alarm rate. The study concludes that BorderWatch is a practical and extensible approach for intelligent surveillance and has direct applications in border security, critical-infrastructure protection, campus and perimeter monitoring, and smart-city safety systems.

Key Points: Multi-Station Surveillance – Object Detection – Multi-Object Tracking – Cross-Camera Object Correlation – Re-Identification – Border Security – Rule-Based Alerting – Computer Vision – Video Analytics – Trajectory Matching – Situational Awareness – Perimeter Monitoring – Real-Time Processing – Simulation

CHAPTER 1 - INTRODUCTION

1.1 Background Information

BorderWatch focuses on validating the accuracy, reliability, and responsiveness of automated, multi-station visual surveillance for border and perimeter monitoring. The process involves analysing footage from geographically distributed camera stations, designing an object detection and tracking pipeline for each station, and building a correlation engine that links object appearances across stations into a single, continuous movement record. Verification is performed by comparing surveillance coverage and traceability before and after correlation is applied, based on detection accuracy, re-identification accuracy, correlation latency, and alert reliability. This helps ensure dependable situational awareness, consistent policy-based alerting, and efficient use of security personnel in border patrol, critical-infrastructure protection, and perimeter security operations.

BorderWatch highlights the importance of structured, rule-guided correlation logic in turning isolated camera feeds into a coherent picture of movement across a monitored region. In this method, a border or perimeter surveillance network is represented as a set of interconnected camera stations, and a combination of detection models, tracking algorithms, and correlation rules is designed to govern how an object's identity is carried from one station to the next. By automatically detecting, tracking, and correlating objects, unnecessary manual review of hours of disconnected footage can be removed while preserving the essential checkpoints of human verification for high-risk alerts. This approach improves situational awareness, reduces manual monitoring overhead, and supports faster response to suspicious activity. Multi-station surveillance and object correlation is widely applicable in border security, critical-infrastructure protection, large campus monitoring, and smart-city safety systems due to its transparency and extensibility.

1.2 Project Objectives

- To study the concept of intelligent multi-station surveillance and cross-camera object correlation.
- To reduce manual monitoring effort while maintaining acceptable detection accuracy and alert reliability.
- To analyse the role of object detection, tracking, and correlation rules in linking movement across stations.
- To compare surveillance effectiveness before and after correlation based on traceability and response time.
- To demonstrate the application of computer vision and rule-based alerting in border security systems.

1.3 Significance of the Study

- Reduces manual video review effort and improves detection of suspicious movement.
- Maintains reliable situational awareness and policy-based alert compliance after automation.
- Demonstrates practical application of computer vision and correlation logic in security systems.
- Improves efficiency in border patrol and critical-infrastructure monitoring operations.

- Supports faster detection, correlation, and alerting of objects of interest.
- Useful in border security, campus security, and smart-city surveillance applications.
- Provides a reliable and computationally efficient technique for cross-station object correlation.

1.4 Scope of the Project

- Applicable in border security, perimeter protection, and critical-infrastructure monitoring systems.
- Can be used for both small, single-corridor deployments and large, multi-station networks.
- Useful in reducing manual review time and improving movement traceability across stations.
- Supports applications in campus security, transport-hub monitoring, and smart-city safety.
- Can be implemented using Python, deep-learning detection models, and video analytics frameworks.
- Helps improve situational visibility and accountability across a distributed camera network.
- Provides a foundation for advanced multi-camera tracking and re-identification techniques.
- Can be extended for real-time alerting and adaptive correlation-rule management.

1.5 Methodology Overview

The methodology of the project begins with collecting multi-station camera footage and modelling the surveillance network as a set of stations connected by expected transit paths. An object detection and tracking pipeline is then implemented at each station to localize and follow people and vehicles within a single camera's field of view. A correlation engine is designed to match tracked objects across stations using appearance embeddings, trajectory continuity, and timing constraints, and alert rules are applied to flag unauthorized or unusual crossing patterns. Using the correlated multi-station data, surveillance performance is evaluated and compared with a station-isolated baseline to assess detection accuracy, re-identification accuracy, and correlation latency. The entire process is implemented and analysed using programming tools such as Python to study the effectiveness of multi-station object correlation.

The project methodology also includes preprocessing and performance-evaluation stages to ensure effective correlation. Initially, the input video streams are sampled, calibrated, and structured into per-station detection and track logs for easier analysis. After applying the correlation engine, different matching thresholds and feature weightings are tested to observe variations in re-identification accuracy and false-match rate. The correlated results are analysed using parameters such as identity-switch rate, correlation latency, and alert precision. Graphical comparisons between station-isolated monitoring and correlated multi-station monitoring are performed to demonstrate the improvement in traceability. This methodology helps in understanding the practical implementation of cross-camera correlation logic in border and perimeter surveillance.

Furthermore, the proposed methodology emphasizes adaptive correlation during system operation by prioritizing the most discriminative features based on lighting, camera angle, and object type. This adaptive selection improves re-identification accuracy without significantly increasing false-match rate. The

methodology also supports faster alert generation and efficient use of security personnel, making it suitable for real-time applications such as border checkpoints, perimeter fences, transport hubs, and wide-area critical-infrastructure sites.

CHAPTER 2 – PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Border and perimeter surveillance systems often experience significant gaps in situational awareness due to manual, station-isolated monitoring, unclear handoff of an object's identity between camera stations, and inconsistent alerting, especially in deployments with many geographically distributed stations. Traditional CCTV-based monitoring often leads to fragmented footage, missed correlations, and slower incident response. The main problem is to reduce the manual effort and delay involved in tracing an object's movement across multiple stations while preserving detection accuracy and alert reliability.

Efficient correlation techniques are necessary to minimize the time taken to link an object's appearance across stations, reduce operator workload, and improve overall situational awareness. This project addresses the problem by using an intelligent multi-station surveillance and object correlation system, a computer-vision technique that reduces monitoring gaps by automatically detecting, tracking, and correlating objects and removing unnecessary manual video review.

2.2 Evidence of the Problem

The rapid growth of border-length camera deployments, remote checkpoints, and wide-area critical-infrastructure sites has increased the demand for efficient, automated video analytics and cross-camera correlation systems. Large, multi-station networks accumulate blind spots and correlation delays at handoff points between stations, requiring greater operator effort and leading to slower incident detection and increased security risk. In many deployments, traditional monitoring methods may either fail to detect cross-station movement in time or provide insufficient automation to correlate it. Studies in video surveillance and computer vision show that a large proportion of monitoring delay is caused by avoidable manual review, which can be reduced using automated detection and correlation techniques. Experimental analysis also demonstrates that cross-station correlation can effectively decrease the time needed to trace an object's movement while maintaining acceptable detection accuracy, proving the existence and importance of the problem.

Multi-station surveillance networks consume significant operator effort and require clear accountability for timely incident detection. With the rapid growth of border security requirements, remote-site monitoring, and wide-area infrastructure protection, efficient object correlation has become essential. Traditional monitoring methods often face problems such as delayed cross-station identification, high manual review overhead, and inconsistent alert prioritization.

2.3 Stakeholders

- Security Engineers – Design and implement the detection, tracking, and correlation pipeline for efficient cross-station monitoring.
- Border Security Agencies – Use correlated surveillance data to reduce blind spots and improve response effectiveness.
- Security Operations Teams – Apply multi-station correlation in control-room monitoring for faster incident resolution.
- Researchers and Academic Institutions – Study and improve object detection, tracking, and re-identification techniques.
- Surveillance Software Providers – Manage correlation engines to optimize multi-camera analytics and reduce operational costs.
- Field Patrol Teams – Coordinate response efficiently through automated alerts with reduced manual review.
- Software Developers – Develop applications and tools for correlation-rule design and alert analysis.
- End Users / Operators – Review flagged incidents efficiently with reduced monitoring workload and clearer prioritization.

2.4 Supporting Data and Research

Research studies in computer vision and video surveillance show that multi-camera object detection and correlation techniques are essential for reducing monitoring gaps and improving traceability. Appearance-based re-identification and trajectory-matching methods have been widely used as effective approaches for linking object identities across non-overlapping camera views while maintaining acceptable accuracy. Experimental results from previous studies indicate that applying well-designed correlation thresholds and alert rules can significantly reduce false alarms with minimal loss of detection sensitivity. Comparative analysis between station-isolated and correlated multi-station monitoring demonstrates improvements in traceability, response time, and operator efficiency. Applications in border security, transport-hub monitoring, campus security, and critical-infrastructure protection further support the reliability and effectiveness of intelligent multi-station correlation techniques.

2.5 Problem Analysis

The main problem in multi-station surveillance is the loss of an object's identity as it moves between camera stations, caused by manual, station-isolated review, unclear correlation logic, and inconsistent alerting. As the number of stations and the length of the monitored border or perimeter increase, the amount of coordination effort required also increases, leading to slower incident detection and higher operational risk. Traditional monitoring methods may fail to detect cross-station movement in time or provide insufficient automation to correlate it. Therefore, an effective technique is needed to minimize avoidable monitoring gaps while preserving detection accuracy and operator oversight for critical alerts.

BorderWatch provides a practical solution to this problem by combining per-station object detection and tracking with a cross-station correlation engine that links object identities using appearance, trajectory, and timing evidence. This reduces the manual effort required to trace movement across stations while

maintaining acceptable detection accuracy and alert reliability. The analysis focuses on evaluating detection accuracy, re-identification accuracy, correlation latency, and false-alarm rate to determine the effectiveness of the proposed approach in multi-station surveillance.

CHAPTER 3 - SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The development and design process of the project begins with selecting suitable multi-station camera footage and surveillance logs for correlation analysis. The surveillance network is modelled as a graph of stations and expected transit paths so that quantitative analysis can be applied easily. A detection and tracking pipeline is then implemented at each station to localize and follow objects of interest within its field of view. The station pairs with the highest cross-traffic and correlation difficulty are identified and targeted with refined matching rules, while low-traffic station pairs are left largely unchanged to achieve optimization without unnecessary computational overhead. After the correlation stage, the surveillance pipeline is re-executed using the correlation engine and compared with the station-isolated baseline to evaluate accuracy and reliability. Different correlation configurations are tested by varying matching thresholds and feature weightings. Performance parameters such as detection accuracy, re-identification accuracy, correlation latency, and false-alarm rate are analysed. The complete system is designed and implemented using programming tools like Python to demonstrate efficient multi-station surveillance using object detection and correlation techniques.

The development and design process of the BorderWatch approach provides an efficient solution for reducing monitoring gaps while maintaining acceptable detection accuracy and operator oversight. The method improves traceability, response time, and overall situational awareness compared to purely manual, station-isolated approaches. It focuses on achieving a balance between automation and human verification by dynamically applying the most relevant correlation rules during system operation. The adaptive nature of the technique allows the system to preserve important controls such as human review of high-risk alerts, evidentiary logging, and privacy safeguards while eliminating unnecessary manual review time. This makes BorderWatch highly suitable for applications like border checkpoints, perimeter fencing, transport-hub security, and wide-area critical-infrastructure monitoring, where both detection accuracy and operational efficiency are important.

3.2 Tools and Technologies Used

- Python – Used for implementing the detection, tracking, and correlation pipeline.
- Deep-Learning Object Detection Models (e.g., YOLO-family architectures) – Used for real-time detection of people and vehicles at each station.
- Multi-Object Tracking Algorithms (e.g., Kalman-filter and appearance-based trackers) – Used for maintaining object identity within a single camera view.
- Appearance Embedding / Re-Identification Models – Used for matching object appearance across stations.
- OpenCV – Performs video frame processing, calibration, and visualization.
- Pandas and NumPy – Perform detection-log analysis and correlation calculations.
- Matplotlib – Generates graphs and visual comparisons between station-isolated and correlated monitoring.

- Multi-Station Camera Dataset / Simulated Footage – Used as input for correlation and evaluation.
- Rule-Based Alerting Engine – Applied for automated detection of unauthorized or unusual crossing patterns.
- Computer System – Used for executing pipelines and evaluating correlation performance.

3.3 Solution Overview

The proposed solution uses an object detection and cross-station correlation pipeline to improve multi-station surveillance by reducing avoidable monitoring gaps while preserving important human oversight controls. In this method, the surveillance network is first modelled as a set of stations, expected transit paths, and timing windows. A detection and tracking module is applied at each station, and a correlation engine matches tracked objects across stations using appearance, trajectory, and timing evidence. The station pairs with the highest correlation difficulty are targeted, while well-performing pairs remain largely unaffected. The correlated surveillance pipeline is then re-executed, resulting in improved traceability and faster alert generation.

For Example

1. Select a multi-station camera dataset and model it as a station-and-path network.
2. Implement a detection and tracking pipeline for object localization, and appearance/trajectory-based correlation logic across stations.
3. Identify high-difficulty station pairs and apply refined matching rules.
4. Re-execute the surveillance pipeline using the correlation-enabled configuration.
5. Compare station-isolated and correlated monitoring based on traceability and alert accuracy.

3.4 Engineering Standards

- Follows IEEE standards for video surveillance systems and intelligent transportation / security automation techniques.
- Ensures reliable object detection and correlation with reduced processing delay according to IEEE video-analytics principles.
- Maintains detection accuracy and alert reliability based on IEEE software engineering guidelines.
- Supports efficient computational methods using standardized detection, tracking, and re-identification techniques.
- Provides secure and privacy-conscious handling of surveillance data in organizational systems.
- Complies with IEEE standards for performance evaluation and system reliability.
- Enhances interoperability with modern security infrastructure and video-management technologies.
- Supports scalable and efficient multi-station surveillance methods for engineering applications.

3.5 Solution Justification

The proposed solution using intelligent multi-station surveillance and object correlation is justified because it provides an efficient method for reducing monitoring gaps while preserving important human oversight controls. The correlation engine removes avoidable manual review by automatically detecting, tracking, and correlating objects across stations, which helps achieve effective situational awareness with minimal disruption to operator verification of critical alerts. Compared to purely manual, station-isolated monitoring, the correlation-based approach offers better consistency and traceability in border and perimeter surveillance applications.

The solution also improves alert response time and reduces operator workload, making it suitable for border checkpoints, perimeter fencing, transport-hub security, and critical-infrastructure protection. Since the approach is based on clearly defined detection and correlation logic, it provides reliable and computationally efficient surveillance. The method can be easily implemented using programming tools such as Python and standard computer-vision libraries, which further supports its practical application in engineering and security-systems management.

CHAPTER 4 – RESULTS AND RECOMMENDATION

4.1 Evaluation of Result

The evaluation of the results shows that intelligent multi-station surveillance and object correlation provide efficient reduction of monitoring gaps with a good balance between automation and human oversight. The correlated surveillance pipeline retained all essential review checkpoints for high-risk alerts even after reducing the manual effort needed to trace objects across stations. Accuracy analysis confirmed that a significant reduction in cross-station identification time was achieved while maintaining acceptable detection precision and recall.

The performance evaluation also indicated that reduced identity-switch rate and improved correlation latency enhance overall situational awareness. Comparison between station-isolated and correlated monitoring showed minimal disruption to operator oversight at moderate correlation-strictness levels. However, overly aggressive correlation thresholds slightly increased false-match rate due to rigid matching criteria under difficult lighting or occlusion conditions. Overall, the evaluation proves that the proposed correlation approach is an effective and reliable technique for multi-station surveillance.

4.2 Challenges Encountered

Several challenges were encountered during the implementation of BorderWatch. One major challenge was maintaining detection accuracy while achieving significant reduction in manual review, as excessive automation of correlation matching caused occasional identity switches under poor lighting, occlusion, or camera-angle variation. Selecting the optimal correlation thresholds for appearance and trajectory matching was also difficult because it directly affected both re-identification accuracy and false-alarm rate. Large, multi-station networks required greater computational effort, data management, and processing time during detection, tracking, and correlation. Additionally, balancing automation with operator oversight and minimizing incorrect correlations were important technical challenges. Efficient implementation of the detection and correlation pipeline using tools such as Python was necessary to improve processing performance and overall system reliability.

4.3 Possible Improvements

- Improve correlation accuracy by selecting optimized matching thresholds dynamically.
- Reduce processing time using lighter and faster detection and re-identification models.
- Implement real-time cross-station alerting for border and perimeter security applications.
- Enhance correlation efficiency for large, wide-area multi-station networks.
- Integrate additional artificial intelligence techniques for adaptive correlation-rule generation.
- Reduce incorrect correlations and improve re-identification accuracy under difficult lighting.
- Extend the method for cross-agency coordination and multi-jurisdiction surveillance networks.
- Optimize resource allocation during detection and correlation pipeline execution.
- Develop user-friendly operator dashboards for easier alert review and rule configuration.

- Apply hybrid techniques by combining rule-based correlation logic with deep-learning re-identification methods.

4.4 Recommendation

It is recommended to use optimized correlation-threshold selection methods to achieve a better balance between traceability and false-alarm reduction. Efficient programming techniques and advanced detection and re-identification algorithms should be implemented to reduce computational time and data-management overhead, especially for large, multi-station networks. The proposed BorderWatch method can be widely applied in border security, transport-hub monitoring, campus security, and critical-infrastructure protection applications to improve situational awareness and response time. Combining rule-based correlation with other modern deep-learning techniques may further enhance performance and reduce incorrect correlations.

CHAPTER 5 – REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

The project provided a clear understanding of multi-station surveillance techniques using object detection, tracking, and cross-camera correlation, and demonstrated the practical application of computer vision in security systems. It helped in learning surveillance-network modelling concepts, detection and tracking pipeline design, correlation-rule design, and the relationship between automation and operator oversight. The project also improved technical skills in using tools such as Python, OpenCV, and deep-learning detection frameworks for implementing video-analytics algorithms. Additionally, it enhanced analytical thinking, problem-solving abilities, and knowledge about detection accuracy, re-identification accuracy, and real-world applications of multi-station surveillance in border security, transport-hub monitoring, and campus safety systems.

5.2 Challenges Encountered and Overcome

Several challenges were encountered during the development of BorderWatch. One of the major difficulties was maintaining a balance between high automation and operator oversight, as overly aggressive correlation matching caused occasional false correlations under difficult conditions. Processing large, multi-station networks required greater computational effort, data management, and longer execution time during detection, tracking, and correlation. Selecting the appropriate matching thresholds for efficient re-identification was also a complex task because it directly influenced correlation accuracy and false-alarm rate. In addition, implementing and optimizing the detection and correlation pipeline using tools such as Python required careful handling of video data and performance-evaluation techniques.

5.3 Application of Engineering Standards

The project applies engineering standards by following established principles of video surveillance systems, computer vision, and computational efficiency. The detection and correlation pipeline is implemented using standard object-detection and re-identification techniques to ensure accurate tracking and reliable cross-station correlation. The system maintains detection accuracy, data integrity, and efficient resource optimization while reducing manual monitoring effort. IEEE-based engineering practices are considered for performance evaluation, algorithm reliability, and efficient handling of surveillance data. The project also follows standard software implementation methods using tools such as Python to ensure accuracy, scalability, and practical applicability in border security and perimeter-monitoring systems.

5.4 Insight into Industry

The project provides valuable insights into the growing importance of intelligent video-analytics technologies in industries such as border security, critical-infrastructure protection, transport-hub monitoring, campus security, and smart-city safety. Industries require efficient correlation methods to manage large volumes of multi-camera video data while reducing operational costs and monitoring gaps. The use of multi-station object correlation demonstrates how computer vision and rule-based alerting

techniques are applied in real-world engineering applications to improve situational awareness and organizational coordination. The project also highlights the increasing demand for advanced surveillance automation solutions in public safety, transportation security, and enterprise-facility protection. Additionally, it shows the importance of computational efficiency, detection transparency, and reliable alert delivery in modern security environments.

5.5 Conclusion of Personal Development

The project contributed significantly to personal and professional development by improving technical knowledge, analytical thinking, and problem-solving skills in the field of computer vision and video surveillance. Working on BorderWatch enhanced understanding of object detection, multi-object tracking, and cross-camera correlation concepts. The project also provided practical experience in using programming tools such as Python for implementing engineering solutions. In addition, it improved research ability, project management skills, and confidence in handling real-world technical challenges. Overall, the project strengthened both theoretical knowledge and practical engineering skills, supporting future academic and professional growth.

CHAPTER 6 – CONCLUSION

The project on BorderWatch, an intelligent multi-station surveillance and object correlation system, successfully demonstrated the effectiveness of combining object detection, tracking, and cross-camera correlation in reducing monitoring gaps while maintaining acceptable operator oversight. The study showed that a well-designed correlation pipeline can efficiently link an object's movement across geographically distributed stations by automatically applying detection, tracking, and matching logic, resulting in improved traceability and faster alert generation. The correlated surveillance pipeline maintained good detection accuracy, proving the reliability of the proposed technique. This project also highlighted the practical applications of multi-station correlation in border security, critical-infrastructure protection, transport-hub monitoring, and campus safety systems. Through implementation and performance analysis, the project provided a clear understanding of surveillance-network modelling, correlation-engine design, and computational efficiency. Overall, the results confirm that BorderWatch is a reliable, efficient, and practically powerful method for multi-station surveillance and real-world security engineering applications.

The BorderWatch approach provides an effective and practically strong technique for reducing monitoring gaps while preserving essential operator oversight. The study confirms that by designing appropriate correlation thresholds, it is possible to achieve a good balance between automation efficiency and detection reliability. This method significantly reduces the time needed to trace an object's movement and improves overall situational awareness, making it highly suitable for modern border and perimeter security applications. Furthermore, the project demonstrates the practical relevance of computer vision in security engineering problem-solving and highlights the potential of correlation-based systems for future advancements in intelligent surveillance. The developed system also helps reduce operator workload, coordination overhead, and response delay, making it suitable for modern multi-station deployments. Performance measures such as Detection Accuracy, Re-Identification Accuracy, and Correlation Latency demonstrate the effectiveness of the proposed approach. Overall, BorderWatch offers a reliable, accurate, and efficient method for intelligent multi-station surveillance and object correlation.

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